

The Iapetus's ridge: Possible explanations of its origin

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Abstract

Images from the Cassini mission show the existence of a long ridge on Iapetus. It extends at least 1400 km along the great circle defined by the equator. We discuss the possibility that the ridge is a result of extensional forces acting above an ascending current of solid-state convection. A two-cell pattern of convection is a reasonable explanation of the observed feature. Three scenarios of the ridge formation are proposed: spin–orbit resonance scenario, convection in low viscosity-interior scenario, and impact generating flow scenario.

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1. Introduction

Iapetus, a medium size icy satellite (MIS), is considered in this paper. The lower limit of MIS radii is approximately 200 km, which corresponds to the smallest spherical icy bodies (e.g. Czechowski, 2006). The upper limit, some 800 km, corresponds to the bodies where water ice is mostly in the form of low-pressure polymorph (ice I). Discussion of the stratification of water ice phases inside the icy satellites is given by Leliwa-Kopystyński et al. (1994). Discussion concerns both for non-differentiated and differentiated models of the satellites interiors. Fourteen bodies (6 satellites of Saturn, 5 of Uranus, 2 of Neptune, and 1 of Pluto) belong to the MIS class.

The authors of this paper considered a possibility of internal convection within the MIS in a few previous papers (e.g. Czechowski and Leliwa-Kopystyński, 2002a,b, 2003, 2005 later on cited as Paper I). The new data provided by the Cassini mission in January 2005 prompted

us to consider the possible scenarios of the evolution of Iapetus.

According to Thomas et al. (2006) the physical data of Iapetus are: mean radius $d = (736.0 \pm 2.0)$ km and mean density $\rho = (1081 \pm 18)$ kg m⁻³. The axes of Iapetus' ellipsoid are $a = 747.1$ km (the Saturn-facing), $b = 749.0$ km (the orbit-facing) and $c = 712.6$ km (polar). Note two unusual facts (i) the longest dimension is parallel to the orbit but it is not Saturn-facing; (ii) the shape of Iapetus corresponds to a spin period of ~ 17 h, instead of the present value 79 days. We believe that the first fact indicates for the large-scale non-homogeneities of density.

Some physical data, as well as the mean orbital elements referred to the local Laplace plane (according to http://ssd.jpl.nasa.gov/sat_elem.html, updated November 30, 2005)¹ are given in Table 1. The relatively large size of Iapetus makes it unlikely that its interior has retained any porosity that it may have inherited from the time of its formation. The evolution of porosity in the icy bodies is discussed by Leliwa-Kopystyński and Kossacki (2000);

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¹ http://ssd.jpl.nasa.gov/sat_elem.html; http://ssd.jpl.nasa.gov/?sat_phys_par; <http://saturn.jpl.nasa.gov/multimedia/images/image-details.cfm?imageID=1270>.

the MISs. The compressional stresses could give rise to thrust faults or folding. It should be noted that strength of the intact rock is not a crucial parameter of the basic theory of thrust faulting (e.g. Watts, 2001, pp. 253–271; Turcotte and Schubert, 2002, pp. 343–350). It is usually assumed that the medium contains many pre-existing joints and faults. Under the tectonic stresses those pre-existing zones of weakness are activated. The effective strength of the rock layer depends mostly on the coefficient of friction and on the weight of the rock layer. Observations of MIS do not indicate many tectonic structures that could be the results of compression. However it is suggested that a few curvilinear ridges on Dione and curvilinear ridge belts on Enceladus resembling fold belts on the Earth could be attributed to compression (Kargel and Pozio, 1996). Compressional origin is also proposed for the lobate scarps on Mercury (Watters and Robinson, 2003).

The stresses resulting from convection depend on many factors and they could reach even 100 MPa (numerical model of convection driven by radiogenic heating in Dione by Forni et al., 1991). This value exceeds the maximum strength of water ice and therefore they are able to fracture the icy lithosphere both in extensional and in compressional regimes. The mean velocity of convection depends on the Rayleigh number. For Ra of the order of 10^5 a typical value of velocity is of the order of 1 mm yr^{-1} (see Paper I). Such velocity of convective flow could lead to displacement on the surface up to 10 km in a period as short as 10^7 years.

Presently available information about Iapetus's ridge is quite limited. However, it seems to be symmetric in respect to its axis and evidently is not curvilinear (it almost follows a great circle). The total length of the ridge is not known but currently documented length is 1400 km therefore one third of the Iapetus circumference (Porco et al., 2005). Moreover, according to Denk et al. (2005): "The ridge itself appears to be located on a broader bulge, extending a few hundred km in southern and northern directions." All these facts fit the hypotheses that the Iapetus's ridge is a structure analogous to the terrestrial ridges frozen in the initial stage of development (compare e.g. Tarbuck and Lutgens, 1988, p. 166; Watts, 2001, p. 286). Therefore we suggest that the ridge is a result of an ancient episode of initial stage of extension. The bulge could be a result of an upward convective current developed below the lithosphere; the ridge itself could be a result of penetration of the lithosphere by cryo-magmas. The similarity of tectonic processes on the Earth and on the icy satellites is not especially surprising. There are other similarities. Giese et al. (2006) state about Ithaca Chasma on Tethys that: "The most striking feature of the Chasma is the upraised flank topography that is elevated up to 6 km above the surroundings. Flank-uplift extends over about 60 km across

strike direction." and "upraised flank topography is typical for rifts on Earth and was also found for rifts on the icy surface of Ganymede." Let us note that Ithaca Chasma extends along great circle like the Iapetus's ridge does. We suggest that the Iapetus's ridge and Ithaca Chasma could be the structures of similar origin. The evolution of the Iapetus's ridge had stopped at an early stage while Ithaca Chasma was developing to more mature stage forming a tectonic valley. Of course, this hypothesis may be revised when more details of the Iapetus's surface will be revealed.

4. Convective patterns

According to McKinnon (1998): "Convection was probable (if not vigorous) in all but the smallest middle size icy satellites early in solar system history." Iapetus is evidently not the smallest one of the MIS; in fact it is one of the largest of MIS. Numerical calculations with temperature-dependent viscosity given by Forni et al. (1991) indicate that convection is possible in Dione. The latter is smaller than Iapetus. Note that the Rayleigh number is proportional to the fifth power of the satellite radius; see Eq. (1). So, if convection is possible in Dione then it will be possible in Iapetus as well.

The patterns of convection driven by tidal heating and of convection driven by exclusively radiogenic heating are generally similar for the range of Rayleigh numbers appropriate for the MISs (Paper I). The patterns are approximately axially symmetric. Two patterns are possible: One-cell pattern and two-cell pattern, Fig. 2. The one-cell pattern leads to bipolar asymmetry (dichotomy between the tectonics of the two hemispheres of the satellite). In the region above an upward flow, extensional surface features and volcanic activity are expected while in the region above a downward flow compressional surface features are expected. A more complicated case is the two-cell convection: there are two regions of compressional tectonics separated by a belt of extensional tectonics. In that case the most important tectonic feature that could accompany convection is a belt of extensional tectonics ridge extending

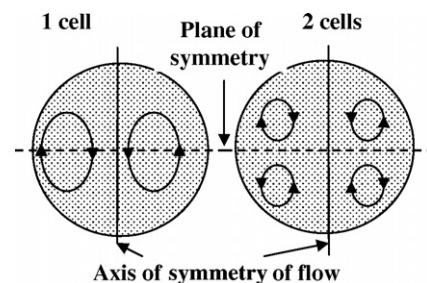


Fig. 2. Two basic patterns of convection characteristic for low and moderate Rayleigh numbers: one-cell pattern and two-cell pattern. The convective patterns have a plane of symmetry ('equatorial region') perpendicular to the axis of symmetry. In reality the symmetry of the convective pattern is not perfect.

over the circumference of the satellite. So, the two-cell pattern seems to be more probably responsible for the origin of the Iapetus's ridge than one-cell pattern. The ridge is formed above the region of extensional tectonics. Note that the surface manifestation of this convection pattern could be masked by the thick, resistant lithosphere or could be damaged by impacts.

Radioactivity and other spherically symmetric heat sources do not prefer any orientation of the convective pattern. Note that rotation of the satellite has negligible influence on the orientation of the convective pattern because the velocity of solid-state convection is very low (of the order of 1 mm per year). However, tidal heating is not spherically symmetric and it leads to convective pattern strongly oriented with respect to the planet–satellite line. Let us consider the two-cell pattern. If the tidal heating is a result of eccentricity of the satellite orbit then the region of extensional tectonics lies in the plane perpendicular to the planet–satellite line. The axis of symmetry of convection should be oriented towards the planet. The ridge would follow the great circle formed by the leading and trailing meridians. If the tidal heating is a result of inclination of the orbit then the region of extensional tectonics lies in the plane of the planet–satellite line and the satellite's axis of rotation. The present orientation of the Iapetus's ridge (in its equatorial plane) does not fit any of the orientations mentioned above. However, this fact could be explained as a result of mass redistribution inside the satellite. The mass redistribution resulting from convection could lead to instability and to reorientation of the satellite. Note that the reorientation is not a cataclysmic event. Therefore, we suggest that after a relatively short period of internal activity Iapetus had changed orientation with respect to the planet–satellite line reaching the present position.

Let us also note that the tidal bulge resulting from a tidal interaction with the planet is usually more massive than most of the surface features (e.g. the ridge) or non-hydrostatic density non-homogeneities (e.g. resulting from convection) in the satellite interior. However, the bulge does not prefer any satellite orientation and does not preclude changes of orientation. If the density redistribution resulting from origin of a ridge and/or convection would prefer change satellite orientation then the tidal bulge will move according to changing orientation of a satellite.

5. Internal heat sources

There are a few potential heat sources in the MIS. They are: primary heat of accretion, self-compaction energy, radiogenic heat, energy of differentiation, energy of phase changes, and tidal heating. All these sources were not intensive enough to give rise to the long lasting endogenic activity on another Saturnian satellite Rhea (see also tectonic activity parameter in Table 1). It is a little larger and a little more massive than Iapetus. Note that the efficiency of the primary heating due to accretion, self-com-

paction energy, energy of differentiation, energy of phase changes is higher for more massive bodies. The radiogenic heating rate is also higher in Rhea than in Iapetus (see Table 1). Therefore, it is unlikely that the mentioned sources, (independent of the satellite orbit), could be responsible for the endogenic activity on Iapetus after the HBE.

Aluminum Al^{26} , could be a powerful source of energy (e.g. Prialnik and Bar-Nun, 1990). The assumption that during formation of the Solar System bodies the Al^{26} was present is very convenient for many hypotheses requiring a substantial heat input. However, the Iapetus's ridge seems to be a structure a few hundreds Myr younger and therefore we cannot treat the ridge as a direct consequence of Al^{26} heating. Of course we do not reject the possibility that Al^{26} decay could be important for thermal history of Iapetus (as well as of other MIS). In this paper we do not discuss scenario of Iapetus evolution supported by presence of Al^{26} .

Let us consider the tidal heating, for Rhea and Iapetus. Using the formula (5) we consider tidal heating for synchronous rotation. Taking the present-day values of the known parameters (ρ, n, d, e) and assuming that the values of material parameters (Q, μ) are equal for both satellites one can calculate the ratio of the tidal heating rates for both bodies. The results presented in Table 1 indicate that the tidal heating rate for Rhea is three orders of magnitude higher than for Iapetus and it is of the same order for Tethys. Therefore Iapetus on its present-day orbit seems to be a worse candidate for a body with tidally driven convection.

6. Evolutionary scenarios

The discussion presented above indicates that Iapetus should not exhibit the large scale traces of the internal global flow. The observation shows the opposite. So, for the interpretation of the Iapetus evolution the special circumstances explaining the effectiveness of the heat sources and/or rheology are necessary. We are looking for a hypothesis explaining the origin of Iapetus's ridge that allows Rhea to be dead. Moreover we would like to avoid of introduction of any *ad hoc* assumptions. To explain the existence of the mysterious ridge on Iapetus we propose three different scenarios. The first scenario is based on the possible spin–orbit resonance, the second scenario is based on the assumption that the viscosity of Iapetus's interior is lower than in the interior of Hera, and the third one is just adaptation of the McKinnon (1985) impact scenario. It seems that all of them are reasonable although the third scenario appears less likely.

6.1. Spin–orbit resonance scenario

We propose the following sequence of events:

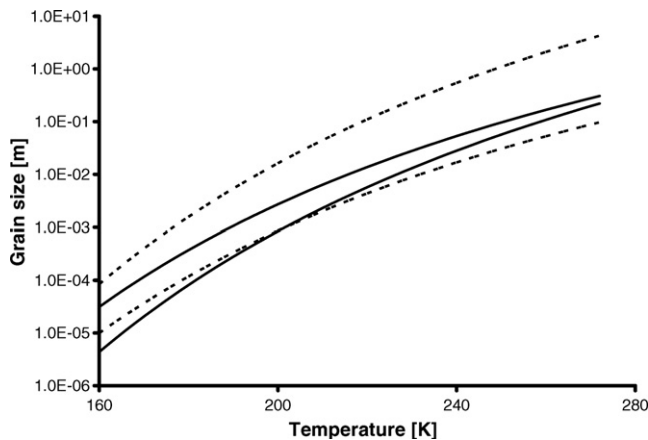


Fig. 4. The grain size versus temperature, according to formula (10) for assumed range of viscosity 10^{19} – 10^{20} Pa s. Upper dotted line is for the grain boundary sliding and for $\eta = 10^{20}$ Pa s. Lower solid line is for the grain boundary sliding and for $\eta = 10^{19}$ Pa s. Upper solid line is for the volume diffusion creep and for $\eta = 10^{20}$ Pa s. Lower dotted line is for the volume diffusion creep and for $\eta = 10^{19}$ Pa s.

Eq. (10) allow us to find some limits for the range of the grain size x versus temperature T , see Fig. 4. The temperature range corresponds, approximately, to that given by Schubert et al. (1986) for thermal models of evolution of Iapetus. The range of the grain size (from 0.01 mm to 10 cm) is realistic. Concluding these considerations we would like to state that the present knowledge about rheology of MISs, does not exclude the possibility of convection. The ammonia content makes the possibility even higher.

6.3. Impact-generated flow scenario

The impact-generated scenario is based on hypothesis of McKinnon (1985) and McKinnon and Benner (1989). They looked for the correlation between an impact event that produced Odysseus crater on Tethys and the formation of the Ithaca Chasma. They suggest that viscous rebound of Odysseus crater could cause the observed fracturing (Ithaca Chasma) by inducing viscous drag at the base of the lithosphere. Recently the mechanism relating the formation of Ithaca Chasma to the Odysseus impact are challenged by the fact that another basin on Tethys (almost as large as Odysseus) did not lead to formation of the structure similar to Ithaca Chasma (Giese et al., 2006). The main difficulty in applying the McKinnon's hypothesis to Iapetus is the lack of a known crater centered on one of the poles of Iapetus. The geographic poles of Iapetus are also the poles of its ridge. Note that the ridge follows a large part of the great circle. The last stage of this scenario is the reorientation of Iapetus into stable, present day position mentioned in scenario 1 (Section 6.1).

7. Conclusions

Let us summarize the results and the proposed scenarios.

- (i) The Iapetus's ridge running along the great circle seems to be a feature of extensional origin. The ridge can be simply interpreted as a result of two-cell convection.
- (ii) The spin–orbit resonance could explain a substantial role of tidal heating in Iapetus. A significant role of this mode of heating is important for our hypothesis concerning origin of the Iapetus's ridge since 2-cell pattern is preferred for the tidal heating. See the scenario 6.1 or combination of scenarios 6.1 and 6.2.
- (iii) To realize the impact generating flow scenario 6.3 a large impact on one of the poles is necessary. Unfortunately the scar of such impact is not identified.
- (iv) The ammonia content in Iapetus is expected. The presence of ammonia supports plausibility of the scenario 6.2. A combination of the scenarios 6.1 and 6.2 seems to be the most plausible.
- (v) The large-scale non-homogeneities of density indicated by present orientation of Iapetus ellipsoid is consistent with our assumption about initial asymmetric shape or/and density distribution in Iapetus.

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